

INTERIM REPORT

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**CHARACTERISTICS OF HUMAN INFLUENZA A H1N1, A H3N2 AND B
VIRUSES ISOLATED OCTOBER 2004 TO JANUARY 2005**

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Of approximately 173 influenza viruses isolated in 24 countries between October 2004 and January 2005, 75% were influenza A, of which 69% were H3N2, and 8% were H1N1. No A H1N2 viruses were received for characterization since February 2004. Influenza B viruses accounted for 23% of viruses.

A H1N1 viruses

In HI tests, using post-infection ferret antisera, the majority of H1N1 viruses were antigenically closely related to the current vaccine strain, A/New Caledonia/20/99 (Table 2).

HA sequences of viruses isolated in Europe and Hong Kong, like many of the sequences obtained by the other CC's, were similar to previous isolates, such as A/England/40/04 (Figure 1) and characterized by the amino acid changes Y252F and V314A (Table 3). None fell within the other phylogenetic group represented by A/Hong Kong/2637/04. HI tests did not consistently distinguish A/Hong Kong/2637/04 from A/New Caledonia/20/99 (Table 2). NA sequences in the two corresponding phylogenetic groups (Figure 2) were similar to those obtained for previous virus isolates (Table 3).

A H3N2 viruses

Most H3N2 viruses were antigenically closely related to the recent reference strains, A/Shantou/1219/04 and A/Oslo/807/04 (using ferret antisera to MDCK grown viruses) and distinguishable from A/Wyoming/3/03 and A/Wellington/1/04 (using antisera to egg grown viruses) (Tables 4A-E). Similar results were obtained with antisera to A/Wyoming/3/03 and A/Wellington/1/04, received from CDC (Table 4B). In contrast, an antiserum to A/Wellington/1/04 from Tokyo was more broadly cross-reactive with recent isolates (Table 4D). Many of the more recent isolates were shown to be antigenically closely related to A/California/7/04 and A/North Dakota/1/04, using antisera received from CDC (Tables 4C-E). Patterns of HI titres were similar for antisera to the recent MDCK and egg grown reference strains (with the K145N change in HA), including A/Singapore/37/04, although the homologous titres varied. (Table 4C and D). A number of viruses gave reduced HI titres with all the reference antisera; however, HA sequences of a number were shown to be similar to the majority of the available H3 sequences and did not indicate the emergence of a distinct antigenic variant.

The majority of HA sequences of recent virus isolates were characterized by the amino acid change K145N and V226I. Two subgroups were distinguished by S227/T326 or P227/K326 (some with a D188N change). Few sequences fell within a phylogenetic group represented by e.g. A/Tokyo/1035/04 or A/Victoria/500/04 and characterized by the changes Y94H, V112I, K145Q, S199P and R269K.

Most NA sequences of recent isolates (e.g. A/Lisbon/3/04) were shown to be similar to that of A/Wellington/1/04 and characterized by the amino acid changes N93D, E199K and Q432E (some with K221E), relative to the NA of A/Fujian/411/02, and were distinguished from the NAs of the phylogenetic group represented by A/Shantou/1219/04 (Figure 4, Table 5).

Influenza B viruses

The majority of B viruses were antigenically closely related to B/Shanghai/361/02 and B/Jiangsu/10/03 (Yamagata lineage). A few, especially from Switzerland, were antigenically closely related to B/Shandong/7/97 and more recent Victoria lineage viruses, such as B/Brisbane/32/02.

Most Yamagata lineage HA sequences of recent isolates fell within the phylogenetic group, represented by B/Oslo/71/04, on the Harbin sublineage, characterized by the amino acid change V252M; some had the additional change P108A (Figure 5, Table 7). Other HAs with V252I had the additional change N123S. A number of isolates had HA sequences which fell within the phylogenetic group represented by B/Jiangsu/10/03 (Figure 5, Table 7). A limited amount of NA sequence data indicated that the NA sequences of the corresponding phylogenetic groups were similar to those of viruses isolated during the previous six months (Table 7).

The HA sequences on the Victoria lineage were similar to those of previous Victoria lineage viruses and close to that of B/Hawaii/13/04 (Table 7). A few viruses, represented by B/Hong Kong/318/04, were distinguished by the changes K80R and K129N. Available NA sequences showed that they were also similar to those of previous Victoria HA/Yamagata NA reassortant viruses.

Acknowledgements

In addition to data received from the other 3 Collaborating Centres, sequence data for viruses isolated during this period were also received from Influenza Centres in Denmark, France, Germany, Hong Kong, Ireland, Israel, Italy, Netherlands, Norway, Portugal, South Africa, Sweden and the UK (indicated by suffixes).

Table 1: Human influenza isolates, October 2004 - present

Country	H1N1	H3N2	B	
			Yamagata lineage	Victoria lineage
October				
China			3	
France		2		
Germany		1		
Hong Kong	2	2		
Ireland	1	1		
Malaysia		1		1
Reunion		1		
Senegal		3		
Spain		1		
Thailand	1			1
November				
Australia		1		
Belgium		2		
China		1	3	
Finland		1		
France		1		
Hong Kong		1		1
Ireland	1			
Norway			1	
Senegal			4	
Spain		1		
Sweden		6		
Switzerland		2		
Turkey			1	
December				
Belgium	1	8		
China		2		
Finland		1		
Germany		9	1	1
Hong Kong		1		
Iceland		8	1	
Latvia		2		
Netherlands	3	3	3	
Norway		4	7	
Spain		10	1	
Sweden		6	1	
Switzerland		3	1	2
Portugal		1	1	
UK	1			
January				
Denmark	2	1		
Egypt				1
Germany		7	2	
Iceland		4		
Norway		3	2	
Portugal	1	6	1	
Spain		5		
Sweden		4		
UK	1	3		
Total =173	14	119	33	7
	6%	69%	19%	4%

Table 2. Antigenic analyses of influenza A H1N1 and H1N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹					
			Post infecton ferret sera					
			A/Beij 262/96	A/NC 20/99	A/Eg 96/02	A/Chile 8885/02	A/HK 2367/04	A/Neth 128/04
A/Beijing/262/1996			640	640	320	320	640	320
A/New Caledonia/20/1999			160	1280	640	640	1280	640
A/Egypt/96/2002			80	640	640	320	640	320
A/Chile/8885/2002			80	640	320	640	640	320
A/Hong Kong/2637/2004			40	640	640	320	640	320
A/Netherlands/128/2004			160	640	640	320	1280	2560
A/Okinawa/42/2004	6.8.04	MDCK4 \1	40	320	320	160	640	640
A/Hong Kong/4555/2004	25.9.04	MDCK2 \1	160	640	1280	640	1280	1280
A/Hong Kong/4580/2004	8.10.04	MDCK2 \1	160	640	1280	640	2560	2560
A/Hong Kong/4596/2004	25.10.04	MDCK2 \1	160	640	1280	640	2560	2560
A/Netherlands/130/2004	Dec-04	MDCK1 \1	80	640	640	320	1280	2560
A/Netherlands/134/2004	Dec-04	MDCK1 \1	160	640	640	640	1280	2560
A/Netherlands/135/2004	Dec-04	MDCK1 \1	80	640	640	320	1280	2560
A/England/57/2004	23.12.04	C1\SIAT1 \1	160	1280	640	320	1280	2560
A/Dakar/79/2004	27.10.04	MDCK2 \1	160	320	320	160	320	320
A/Dakar/80/2004	28.10.04	MDCK2 \1	160	640	640	160	320	640
A/Norway/69/2004	31.12.04	MDCK1 \1	160	1280	640	320	2560	2560
A/Denmark/3/2005	6.1.05	MDCK2 \1	160	640	640	320	1280	2560
A/England/69/2004	10.1.05	C1\SIAT1 \1	160	640	640	320	1280	2560
A/Denmark/5/2005	11.1.05	MDCK1 \1	160	1280	640	320	1280	2560
A/Lisbon/17/2005	12.1.05	MDCK1 \1	160	640	640	320	1280	2560
A/Bangkok/1544/2004	unknown	E5 \1	160	640	640	320	1280	1280
A/Florida/4/2004	unknown	x1,C2 \1	80	640	640	320	640	1280

1. < : <40

Figure 1. Phylogenetic comparison of H1 HAs

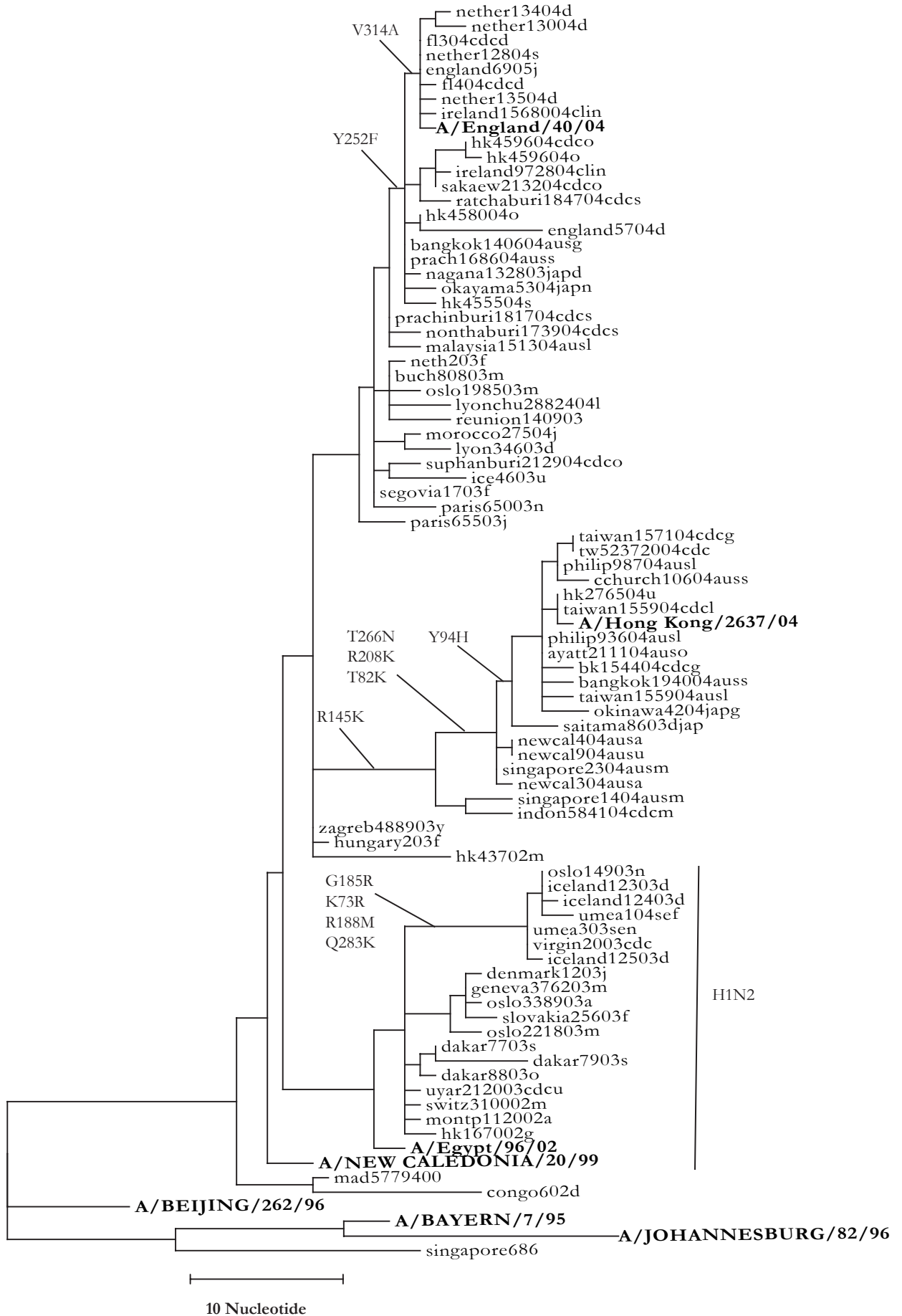


Figure 2. Phylogenetic comparison of N1

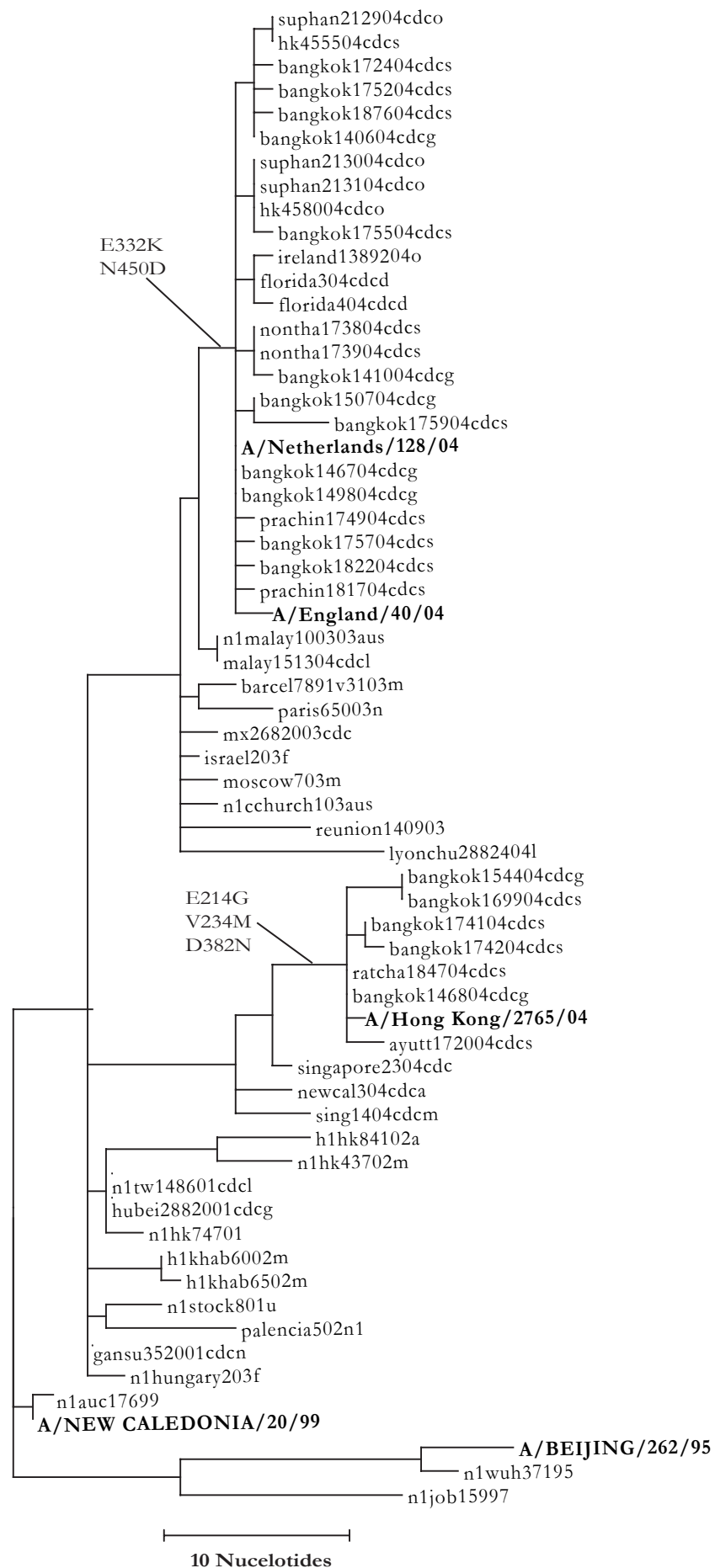


Table 3. Amino acid differences characteristic of HA and NA sequences of H1N1 viruses

Variant Group	Representative strain	Amino acid changes ¹	
		HA	NA
1	A/England/40/04	V166A W251R Y252F V314A	V48I E332K N450D
2	HK/2765/04	T82K Y94H R145K R208K T266N	E214G R222Q V234M D382N

1. Relative to A/New Caledonia/20/99 (H1N1)

Table 4A. Antigenic analyses of influenza A H3N2 viruses (13.1.05)

Viruses			Haemagglutination inhibition titre ¹										
			Post infection ferret sera										
			A/Pan 2007/99 F2/01	A/Wy 3/03 F11/03	A/Chch 28/03 F3/04	A/Well 1/04 F12/04	A/Shan 1219/04 F16/04	A/Oslo 807/04 F17/04	A/Fin 486/04 F22/04	A/Vic 500/04 AUS	A/Sing 37/04 AUS	A/Vic 513/04 AUS	A/Vic 110/04 AUS
A/Panama/2007/1999			5120	640	320	1280	80	160	40	80	160	160	160
A/Wyoming/3/2003			640	5120	5120	2560	640	1280	320	160	640	160	1280
A/Christchurch/28/2003			80	640	1280	320	160	160	80	80	80	80	640
A/Wellington/1/2004			160	1280	1280	5120	1280	1280	320	320	640	640	1280
A/Shantou/1219/2004			80	160	320	320	320	320	160	160	320	160	320
A/Oslo/807/2004			40	160	160	160	160	320	160	160	160	160	320
A/Finland/486/2004			80	160	640	640	1280	1280	640	320	640	320	1280
A/Victoria/500/2004	13.8.04	E6	40	640	640	320	2560	2560	640	640	640	2560	2560
A/Singapore/37/2004	7.6.04	E4	160	640	640	640	2560	2560	1280	1280	2560	2560	2560
A/Victoria/513/2004	30.9.04	E4	40	640	640	320	2560	2560	1280	640	2560	2560	2560
A/Shantou/1386/2004	17.8.04	MDCK1	160	640	640	640	1280	640	320	320	640	640	1280
A/Shantou/1390/2004	20.8.04	MDCK1	80	320	640	160	1280	1280	320	160	320	320	640
A/Shantou/1395/2004	22.8.04	MDCK1	80	320	320	160	1280	1280	320	160	320	320	640
A/Shantou/1401/2004	1.9.04	MDCK1	160	1280	1280	1280	1280	1280	320	320	640	640	1280
A/Shantou/1255/2004	3.9.04	MDCK1	80	80	320	80	1280	1280	160	320	320	320	640
A/Shantou/1263/2004	6.9.04	MDCK1	160	640	1280	640	1280	1280	320	320	640	640	1280
A/Shantou/1273/2004	8.9.04	MDCK1	160	640	640	640	1280	640	160	320	640	640	1280
A/Shantou/1441/2004	8.9.04	MDCK1	80	320	320	320	320	1280	320	160	320	320	640
A/Shantou/1450/2004	11.9.04	MDCK1	80	80	320	80	80	80	80	160	160	160	640
A/Shantou/1308/2004	14.9.04	MDCK1	80	160	160	80	320	320	160	320	320	320	640
A/Shantou/1309/2004	14.9.04	MDCK1	80	640	640	160	1280	1280	320	320	640	640	1280
A/Shantou/1319/2004	15.9.04	MDCK1	40	80	160	80	640	1280	160	160	160	160	640
A/Shantou/1320/2004	15.9.04	MDCK1	80	640	320	160	1280	1280	320	320	640	320	1280
A/Hong Kong/4465/2004	15.9.04	MDCK2	40	160	160	80	160	160	160	160	160	160	160
A/Shantou/1471/2004	22.9.04	MDCK1	40	320	160	160	320	640	160	160	160	160	320
A/Shantou/1478/2004	27.9.04	MDCK1	80	320	320	320	640	1280	320	160	320	320	640
A/Hong Kong/4584/2004	11.10.04	MDCK2	80	320	640	320	1280	1280	320	320	640	640	640
A/Hong Kong/4593/2004	18.10.04	MDCK2	160	320	640	320	1280	1280	320	320	640	320	640
A/Norway/1103/2004	22.11.04	MDCK1	160	640	640	640	1280	2560	640	320	640	640	1280
A/Hong Kong/4607/2004	23.11.04	MDCK2	<	80	160	80	80	160	160	80	80	80	160
A/Madrid/G1681/2004	27.11.04	MDCK2	80	640	1280	640	2560	2560	1280	640	1280	640	1280
A/Madrid/RR1735/2004	Dec-04	MDCKx	40	160	320	80	1280	1280	160	160	160	160	1280
A/Madrid/RR1734/2004	Dec-04	MDCKx	80	640	640	160	1280	1280	640	320	640	320	1280
A/Madrid/G1687/2004	1.12.04	MDCK2	80	160	320	160	1280	1280	160	160	320	160	1280
A/Madrid/RR1731/2004	12.12.04	MDCKx	40	80	160	80	320	640	160	160	160	80	320
A/Madrid/G1697/2004	14.12.04	MDCK2	80	320	320	160	1280	1280	320	320	320	160	1280
A/Latvia/15743/2004	18.12.04	MDCK1	80	320	640	320	1280	1280	320	320	640	320	1280

1. <= <40

Table 4B. Antigenic analyses of influenza A H3N2 viruses (18.1.05)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Pan	A/Wy	A/Wy	A/Chch	A/Well	A/Well	A/Shan	A/Oslo	A/Fin	A/Sing
			2007/99	3/03	3/03	28/03	1/04	1/04	1219/04	807/04	486/04	37/04
			F2/01	F11/03	CDC	F3/04	F13/04	CDC	F16/04	F17/04	F22/04	AUS
A/Panama/2007/1999			5120	320	640	160	160	160	80	40	<	80
A/Wyoming/3/2003			320	2560	5120	2560	320	640	640	640	320	640
A/Christchurch/28/2003			80	1280	2560	2560	160	160	160	160	80	160
A/Wellington/1/2004			320	2560	5120	1280	2560	5120	2560	1280	640	1280
A/Shantou/1219/2004			80	160	320	320	160	160	640	320	160	320
A/Oslo/807/2004			80	160	320	160	160	160	320	320	160	320
A/Finland/486/2004	3.11.04	MDCK2	80	160	640	320	320	320	1280	1280	640	640
A/Singapore/37/2004	7.6.04	E4	160	320	2560	320	320	640	1280	640	640	1280
A/Netherlands/132/2004	Dec-04	MDCK1 \1	80	320	640	320	320	320	320	640	320	640
A/Netherlands/138/2004	Dec-04	MDCK1 \1	160	320	640	640	320	640	640	640	640	640
A/Berlin/35/2004	2.12.04	MDCKx \1	80	320	640	160	160	160	320	320	80	320
A/Hong Kong/4614/2004	3.12.04	MDCK2 \2	40	160	160	80	80	160	160	160	160	160
A/Berlin/37/2004	15.12.04	MDCKx \1	80	160	320	160	160	160	320	320	80	160
A/Lisbon/4/2004	19.12.04	MDCK2 \3	40	160	160	160	160	160	320	160	160	320
A/Tokyo/1035/2004	19.10.04	MDCK3 \1	160	320	640	640	320	640	1280	160	80	640
A/Norway/1191/2004	22.12.04	MDCK1 \1	80	160	320	160	160	160	320	320	160	320
A/England/77/2005	10.1.05	C1\SIAT1 \1	320	1280	1280	1280	640	1280	2560	1280	640	1280

1. <=<40

Table 4C. Antigenic analyses of influenza A H3N2 viruses (21.1.05)

Viruses			Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Pan 2007/99 F2/01	A/Wy 3/03 F11/03	A/Chch 28/03 F3/04	A/Well 1/04 F12/04	A/Shan 1219/04 F16/04	A/Oslo 807/04 F17/04	A/Fin 486/04 F22/04	A/Sing 37/04 AUS	A/Cal 7/04 CDC	A/ND 1/04 CDC
A/Panama/2007/1999			5120	320	320	320	160	320	40	80	160	40
A/Wyoming/3/2003			2560	5120	5120	2560	2560	2560	1280	640	2560	2560
A/Christchurch/28/2003			320	1280	2560	640	640	320	320	160	320	160
A/Wellington/1/2004			320	1280	1280	1280	2560	1280	640	1280	1280	1280
A/Shantou/1219/2004			160	160	320	320	640	320	320	320	640	320
A/Oslo/807/2004			160	40	320	320	1280	1280	640	640	1280	640
A/Finland/486/2004	3.11.04	MDCK2	40	160	640	640	1280	1280	640	640	1280	640
A/Singapore/37/2004	7.6.04	E4	160	160	640	1280	2560	2560	1280	5120	2560	1280
A/California/7/2004		SpfCk3, E3	80	160	1280	640	2560	2560	640	1280	1280	1280
A/North Dakota/1/2004		SpfCk3, E4	80	80	160	160	1280	1280	160	640	640	320
A/Shantou/1307/2004	13.9.04	MDCK1 \2	80	320	160	160	640	320	320	320	640	640
A/Shantou/1465/2004	18.9.04	MDCK1 \2	<	160	80	160	160	160	80	160	320	320
A/Belgium/2/2004	16.11.04	MDCK2 \2	80	160	640	320	1280	1280	640	640	640	640
A/Switzerland/4689/2004	26.11.04	MDCK1 \1	80	320	320	160	640	1280	320	640	640	1280
A/Switzerland/4817/2004	30.11.04	MDCK1 \1	<	160	320	160	160	640	160	160	640	640
A/Valladolid/1/2004	7.12.04	MDCK2 \3	40	160	160	160	320	320	160	320	320	320
A/Berlin/36/2004	9.12.04	MDCKx \2	80	320	320	320	1280	1280	640	1280	640	1280
A/Belgium/4/2004	10.12.04	MDCK2 \2	80	320	320	160	640	640	160	320	640	640
A/Iceland/8/2004	13.12.04	MDCK2 \1	80	320	160	160	640	640	320	320	640	640
A/Berlin/38/2004	15.12.04	MDCKx \2	80	320	160	160	320	640	320	320	640	320
A/Iceland/7/2004	15.12.04	MDCK1 \1	80	320	320	320	1280	1280	320	1280	1280	1280
A/Latvia/15447/2004	18.12.04	MDCK2 \3	80	160	160	160	160	320	160	160	320	320
A/Berlin/39/2004	20.12.04	MDCKx \2	<	160	160	160	320	320	160	160	320	320
A/Berlin/41/2004	22.12.04	MDCKx \2	40	320	160	160	320	640	160	320	640	640
A/Berlin/40/2004	22.12.04	MDCKx \2	<	160	160	160	320	320	160	160	320	320
A/Iceland/12/2004	22.12.04	MDCK2 \1	80	320	320	320	1280	1280	640	640	1280	1280
A/Belgium/138/2004	27.12.04	MDCK1 \2	80	320	320	320	1280	1280	320	640	640	640
A/Belgium/146/2004	28.12.04	MDCK1 \2	40	320	160	160	640	640	160	320	640	640
A/Belgium/147/2004	28.12.04	MDCK1 \2	80	320	320	160	1280	1280	320	640	640	640
A/Iceland/15/2004	28.12.04	MDCK1 \1	40	160	160	160	320	640	160	160	320	640
A/Iceland/18/2004	29.12.04	MDCK1 \1	80	320	160	160	320	640	160	320	640	640
A/Iceland/19/2004	29.12.04	MDCK1 \1	40	160	320	160	1280	1280	160	320	640	640
A/Iceland/22/2004	30.12.04	MDCK1 \1	40	320	160	160	640	640	160	320	640	640
A/Iceland/20/2004	30.12.04	MDCK1 \1	80	320	160	160	320	640	160	320	640	640
A/Iceland/1/2005	1.1.05	MDCK1 \1	80	320	320	320	1280	1280	320	640	1280	1280
A/Iceland/3/2005	4.1.05	MDCK1 \1	80	320	320	320	1280	1280	160	640	640	640
A/Iceland/5/2005	7.1.05	MDCK0 \1	40	160	80	160	160	160	80	160	160	320
A/Iceland/6/2005	7.1.05	MDCK1 \1	80	160	160	160	640	320	160	320	320	320

1. <=40

Table 4D. Antigenic analyses of influenza A H3N2 viruses (25.1.05)

Viruses			Haemagglutination inhibition titre ¹										
			Post infection ferret sera										
			A/Pan 2007/99 F2/01	A/Wy 3/03 F11/03	A/Chch 28/03 F3/04	A/Well 1/04 F12/04	A/WELL 1/04 JAP	A/Shan 1219/04 F16/04	A/Oslo 807/04 F17/04	A/Fin 486/04 F22/04	A/Sing 37/04 AUS	A/Cal 7/04 CDC	A/ND 1/04 CDC
A/Panama/2007/1999			2560	320	160	160	320	40	160	<	80	80	<
A/Wyoming/3/2003			640	5120	5120	1280	2560	1280	1280	640	320	1280	640
A/Christchurch/28/2003			160	1280	2560	320	640	320	320	160	160	320	160
A/Wellington/1/2004			160	1280	1280	2560	2560	2560	1280	640	1280	1280	640
A/Shantou/1219/2004			40	160	320	160	320	640	160	160	320	320	320
A/Oslo/807/2004			40	160	160	160	320	320	320	160	320	640	320
A/Finland/486/2004	3.11.04	MDCK2	40	160	320	160	640	2560	2560	640	640	2560	640
A/Singapore/37/2004	7.6.04	E4	160	640	640	640	2560	2560	2560	2560	5120	2560	2560
A/California/7/2004		SpfCk3, E3	80	1280	2560	320	1280	2560	2560	1280	640	2560	1280
A/North Dakota/1/2004		SpfCk3, E4	80	160	320	160	640	1280	1280	320	640	1280	640
A/Stockholm/15/2004	26.9.04	MDCK1	160	1280	1280	640	1280	2560	2560	640	1280	1280	1280
A/Stockholm/17/2004	Nov-04	MDCK1	80	1280	1280	640	1280	2560	2560	640	1280	2560	1280
A/Belgium/1/2004	13.11.04	MDCK2	80	640	640	320	640	1280	2560	640	640	1280	1280
A/Stockholm/16/2004	13.11.04	MDCK2	80	320	160	320	640	320	640	320	320	640	640
A/Malmo/2/2004	16.12.04	MDCK2	80	320	320	160	320	320	320	160	320	640	640
A/Stockholm/18/2004	19.11.04	MDCK2	80	320	320	160	640	640	1280	320	320	1280	1280
A/Stockholm/19/2004	23.11.04	MDCK1	160	1280	1280	320	640	2560	2560	640	640	1280	1280
A/Stockholm/20/2004	29.11.04	MDCK2	80	320	320	160	640	640	1280	320	320	1280	1280
A/Stockholm/21/2004	Dec-04	MDCK2	80	160	320	80	160	160	160	160	160	160	160
A/Netherlands/140/2004	Dec-04	MDCK1	80	1280	640	640	1280	640	1280	320	640	1280	640
A/Umea/1/2005	7.12.04	MDCK2	80	320	320	320	1280	2560	2560	640	640	1280	1280
A/Umea/2/2005	8.12.04	MDCK2	80	640	320	320	1280	2560	2560	640	640	1280	1280
A/Belgium/8/2004	8.12.04	MDCK2	160	320	640	320	2560	2560	2560	1280	1280	2560	2560
A/Belgium/3/2004	10.12.04	MDCK2	160	640	640	320	640	2560	2560	1280	640	1280	1280
A/Belgium/10/2004	10.12.04	MDCK2	160	1280	1280	320	2560	2560	2560	2560	1280	2560	1280
A/Belgium/9/2004	10.12.04	MDCK2	80	640	320	640	1280	2560	2560	1280	1280	1280	1280
A/Finland/487/2004	21.12.04	MDCK2	80	640	320	640	1280	2560	2560	1280	1280	2560	1280
A/Switzerland/5732/2004	29.12.04	MDCK1	80	320	160	320	1280	640	1280	320	640	1280	1280
A/Norway/44/2005	3.1.05	MDCK1	80	320	320	320	640	320	320	160	320	320	320
A/Stockholm/1/2005	3.1.05	MDCK1	160	1280	1280	320	1280	1280	2560	640	1280	1280	1280
A/Norway/70/2005	4.1.05	MDCK1	80	640	160	640	1280	1280	2560	1280	640	1280	1280
A/England/73/2005	7.1.05	MK1/SIAT1	160	640	640	320	1280	2560	2560	2560	1280	2560	1280

1. <=40

Table 4E. Antigenic analyses of influenza A H3N2 viruses (1.2.05)

Viruses	Isolation Date		Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Pan 2007/99 F2/01	A/Wy 3/03 F11/03	A/Chch 28/03 F3/04	A/Well 1/04 F13/04	A/Shan 1219/04 F16/04	A/Oslo 807/04 F17/04	A/Fin 486/04 F22/04	A/Cal 7/04 2005-025 CDC
A/Panama/2007/1999		Ex	5120	1280	320	320	320	640	40	160
A/Wyoming/3/2003		Ex	1280	5120	5120	1280	5120	5120	1280	2560
A/Christchurch/28/2003		Ex	160	2560	2560	160	320	320	320	320
A/Wellington/1/2004		Ex	320	5120	2560	2560	5120	2560	1280	2560
A/Shantou/1219/2004		MDCKx	80	320	320	320	640	320	160	640
A/Oslo/807/2004		MDCK2	80	320	320	160	640	640	160	640
A/Finland/486/2004	3.11.04	MDCK2	80	160	320	160	1280	2560	640	1280
A/California/7/2004		SpfCk3, E3	80	640	640	320	2560	2560	1280	2560
A/Shantou/1283/2004	9.9.04	MDCK1 \1	160	1280	1280	640	2560	2560	1280	1280
A/Shantou/1296/2004	13.9.04	MDCK1 \1	160	1280	1280	640	2560	2560	1280	1280
A/Dakar/76/2004	20.9.04	MDCK2 \2	80	640	640	320	1280	2560	640	1280
A/Shantou/1333/2004	20.9.04	MDCK1 \1	160	1280	640	320	2560	2560	1280	1280
A/Norway/58/2004	20.12.04	MDCK1 \2	160	1280	1280	320	2560	2560	1280	1280
A/Switzerland/5625/2004	22.12.04	MDCK1 \2	160	1280	1280	320	2560	2560	1280	1280
A/Norway/13/2004	24.12.04	MDCK1 \2	80	320	320	160	640	1280	160	640
A/Nordrhein/33/2004	27.12.04	MDCKx \1	80	320	320	160	640	640	320	1280
A/Barcelona/53/2004	28.12.04	MDCK \1	80	640	640	320	1280	2560	640	640
A/Barcelona/58/2004	28.12.04	MDCK \1	80	320	640	160	1280	2560	640	640
A/Bayern/16/2004	29.12.04	MDCKx \1	80	320	160	160	640	640	320	640
A/Switzerland/5732/2004	29.12.04	MDCK1 \2	40	640	320	80	1280	1280	640	640
A/Barcelona/59/2004	30.12.04	MDCK \1	40	160	320	160	1280	2560	320	640
A/Umea/3/2005	30.12.04	MDCK2 \2	40	640	320	80	1280	2560	640	640
A/Lisbon/16/2005	Jan-05	MDCK1 \1	80	1280	640	160	2560	2560	1280	1280
A/Lisbon/14/2005	2.1.05	MDCK1 \1	80	1280	640	160	2560	2560	1280	1280
A/Berlin/43/2004	4.1.05	MDCKx \1	80	320	320	160	640	1280	640	1280
A/Norway/71/2005	5.1.05	MDCK1 \2	80	640	640	320	1280	2560	640	1280
A/Barcelona/74/2005	5.1.05	MDCK1 \1	40	160	640	80	1280	2560	640	640
A/Stockholm/2/2005	5.1.05	MDCK1 \2	40	320	160	80	640	640	160	640
A/Denmark/2/2005	6.1.05	MDCK2 \1	80	640	320	160	1280	2560	640	1280
A/Lisbon/4/2005	6.1.05	MDCK1 \1	80	1280	640	320	2560	2560	1280	1280
A/Lisbon/6/2005	7.1.05	MDCK2 \1	80	1280	640	160	2560	2560	1280	1280
A/Lisbon/10/2005	9.1.05	MDCK1 \1	80	1280	640	160	2560	2560	1280	1280
A/Barcelona/86/2004	10.1.05	MDCK1 \1	40	160	320	80	1280	1280	320	320
A/Nordrhein/32/2004	10.1.05	MDCKx \1	80	320	320	160	640	1280	320	640
A/Lisbon/12/2005	11.1.05	MDCK2 \1	80	1280	640	160	2560	2560	1280	1280
A/Thuringen/2/2005	12.1.05	MDCKx \1	40	320	320	160	640	1280	320	640
A/Thuringen/1/2005	13.1.05	MDCKx \1	80	320	320	160	640	640	320	640
A/Berlin/1/2005	14.1.05	MDCKx \1	80	320	320	160	640	1280	320	640
A/Barcelona/104/2005	17.1.05	MDCK \1	160	1280	1280	320	2560	2560	1280	1280
A/Barcelona/101/2005	17.1.05	MDCK \1	80	160	320	160	640	2560	320	320
A/Barcelona/99/2005	17.1.05	MDCK \1	40	160	320	80	640	1280	320	640
A/Berlin/3/2005	17.1.05	MDCKx \1	80	320	320	160	640	1280	320	640
A/Berlin/4/2005	20.1.05	MDCKx \1	40	160	160	80	320	640	160	640
A/Tennessee/6/2004	unknown	M2,C1 \1	40	160	80	80	320	320	160	640

1. <=<40

Figure 3. Phylogenetic comparison of H3 HAs

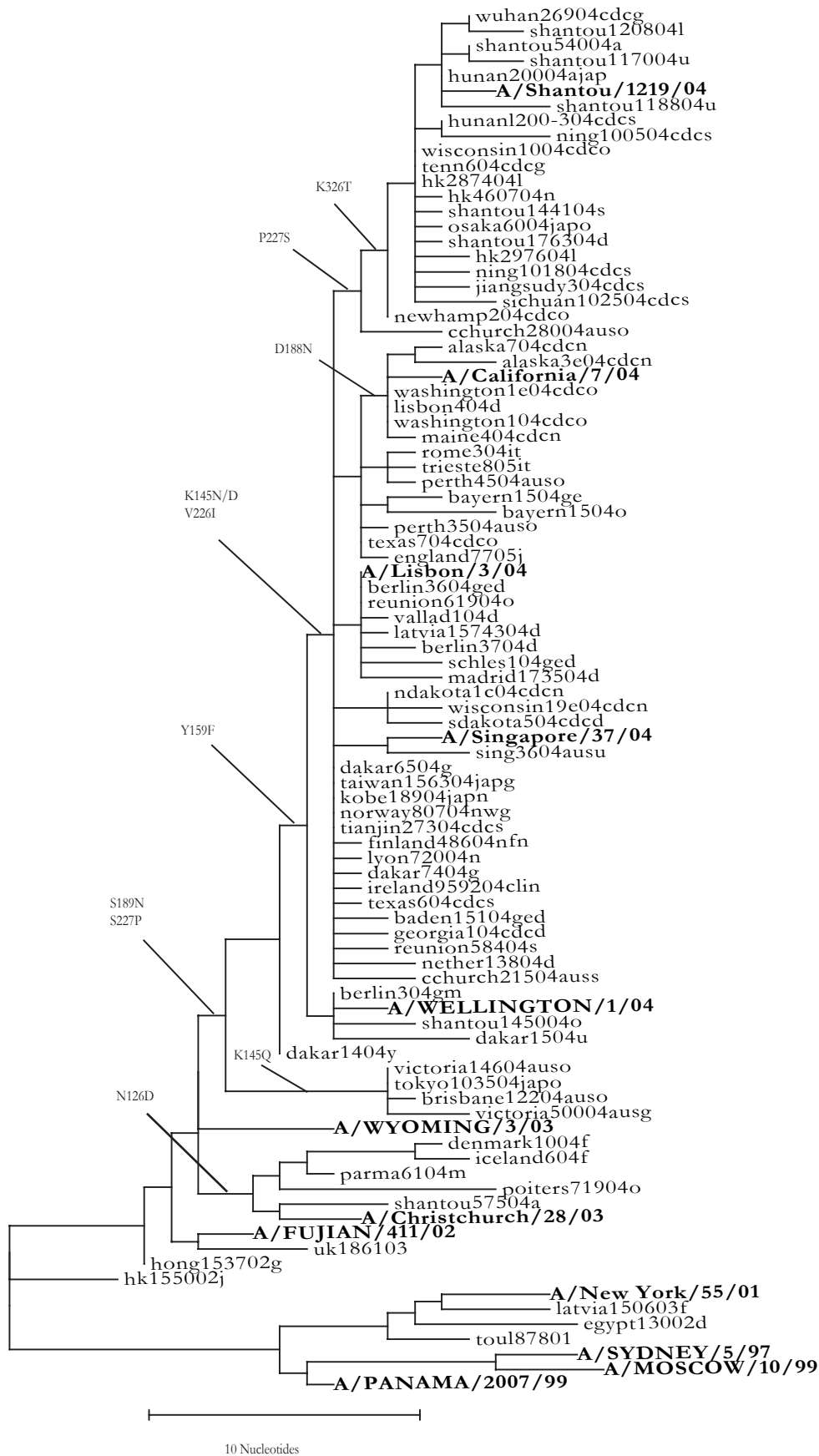


Figure 4. Phylogenetic comparison of N2

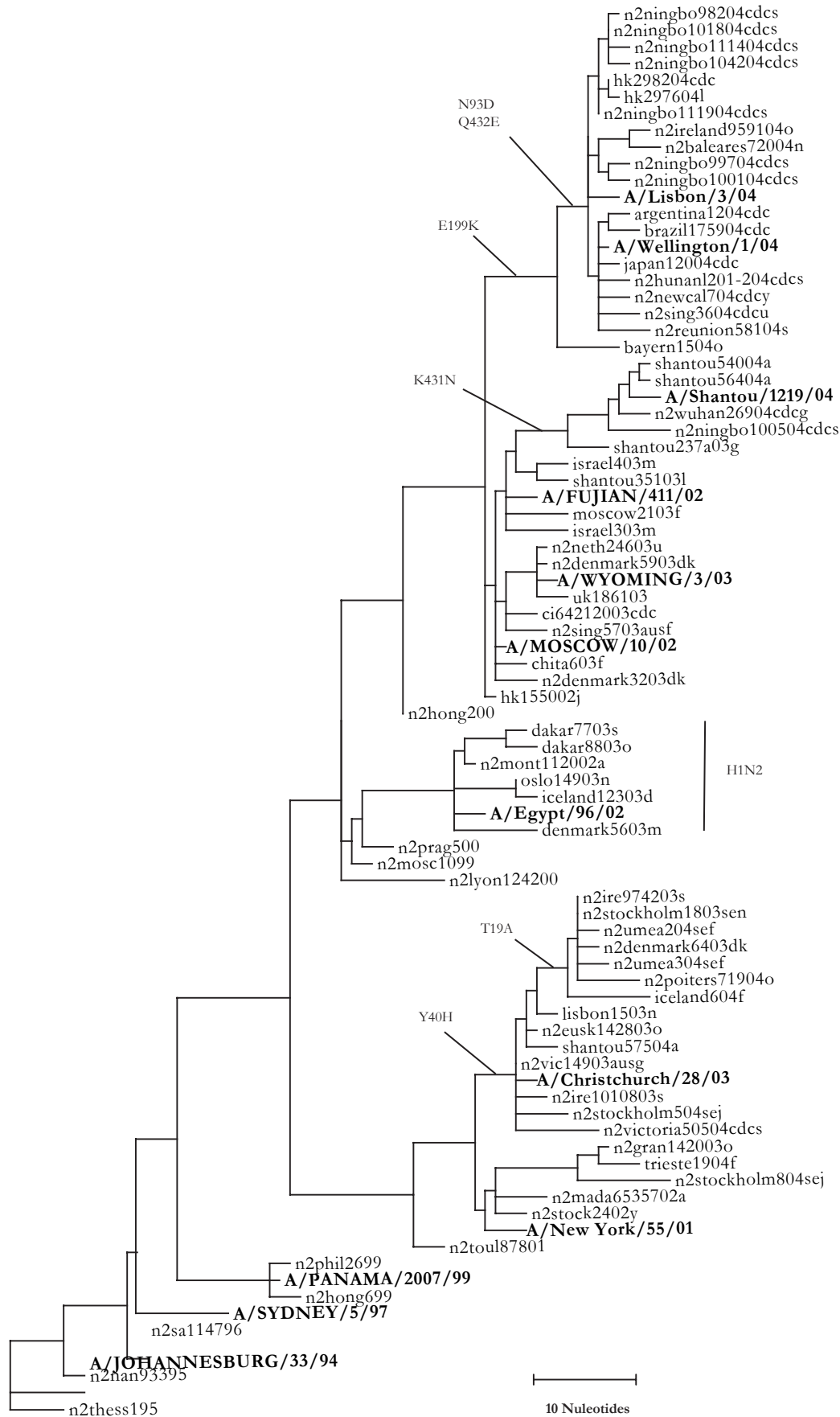


Table 5. Amino acid changes characteristic of H3N2 sequence variants

Variant group	Representative strain	Amino Acid changes ¹	
		HA	NA
1	A/Lisbon/3/04	K145N Y159F S189N V226I S227P	N93D E199K Q432E (K221E)
	A/Shantou/1219/04	K145N Y159F S189N V226I K326T	(E199K) I215V K431N

1. Relative to A/Fujian/411/02.

Table 6A. Antigenic analyses of influenza B viruses (13.1.05)

			Haemagglutination inhibition titre ¹						
Viruses	Isolation date		Post infection ferret sera						
			B/Shan ²	B/Shan	B/Tehr	B/Bris	B/Sich	B/Shai	B/Jiang
			7/97	7/97	80/02	32/02	379/99	361/02	10/03
B/Shandong/7/1997			5120	320	160	160	<	<	<
B/Tehran/80/2002			5120	320	640	320	<	<	<
B/Brisbane/32/2002			2560	160	160	320	<	<	<
B/Sichuan/379/1999			<	<	<	<	320	320	40
B/Shanghai/361/2002			<	<	<	<	320	640	80
B/Jiangsu/10/2003			<	<	<	<	80	80	320
B/Shantou/1410/2004	5.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1257/2004	6.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1271/2004	7.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1435/2004	8.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1294/2004	13.9.04	MDCK1 \1	<	<	<	<	160	320	320
B/Shantou/1343/2004	21.9.04	MDCK1 \1	<	<	<	<	80	160	320
B/Shantou/1468/2004	22.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1469/2004	23.9.04	MDCK1 \1	<	<	<	<	160	160	160
B/Shantou/1475/2004	27.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1476/2004	27.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1480/2004	29.9.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1424/2004	12.10.04	MDCK1 \1	<	<	<	<	80	160	160
B/Shantou/1486/2004	15.10.04	MDCK1 \1	<	<	<	<	160	160	320
B/Shantou/1577/2004	17.10.04	MDCK1 \1	<	<	<	<	80	160	160
B/Shantou/1633/2004	12.11.04	MDCK1 \1	<	<	<	<	80	160	160
B/Shantou/1647/2004	16.11.04	MDCK1 \1	<	<	<	<	80	160	160
B/Shantou/1681/2004	23.11.04	MDCK1 \1	<	<	<	<	80	160	160

¹ <, <40

² hyperimmune sheep serum

Table 6B. Antigenic analyses of influenza B viruses (18.1.05)

Viruses			Haemagglutination inhibition titre ¹							
			Isolation date	B/Shan ² 7/97	Post infection ferret sera					
					B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03
B/Shandong/7/1997			2560	320	160	160	<	<	<	
B/Tehran/80/2002			2560	320	640	320	<	<	<	
B/Brisbane/32/2002			1280	160	160	320	<	<	<	
B/Sichuan/379/1999			<	<	<	<	320	320	<	
B/Shanghai/361/2002			<	<	<	<	320	320	40	
B/Jiangsu/10/2003			<	<	<	<	80	160	320	
B/Netherlands/131/2004	Dec-04	MDCK1 \1	<	<	<	<	160	160	160	
B/Netherlands/133/2004	Dec-04	MDCK1 \1	<	<	<	<	160	320	320	
B/Netherlands/141/2004	Dec-04	MDCK1 \1	<	<	<	<	160	160	160	
B/Madrid/G1695/2004	10.12.04	MDCK2 \2	<	<	<	<	160	160	160	
B/Norway/1201/2004	21.12.04	MDCK1 \1	<	<	<	<	160	160	160	
B/Norway/1209/2004	21.12.04	MDCK1 \1	<	<	<	<	80	160	160	
B/Norway/1192/2004	22.12.04	MDCK1 \1	<	<	<	<	80	160	160	
B/Norway/1189/2004	22.12.04	MDCK1 \1	<	<	<	<	160	160	320	
B/Norway/1190/2004	22.12.04	MDCK1 \1	<	<	<	<	80	160	160	
B/Lisbon/1/2004	22.12.04	MDCK2 \2	<	<	<	<	160	320	320	
B/Norway/1230/2004	28.12.04	MDCK1 \1	<	<	<	<	160	160	320	
B/Hong Kong/318/2004	15.12.04	MDCK2 \1	1280	<	<	<	<	<	<	

¹ <, <40

² hyperimmune sheep serum

Table 6C. Antigenic analyses of influenza B viruses (21.1.05)

Viruses	Isolation date		Haemagglutination inhibition titre ¹							
			B/Shan ² 7/97	Post infection ferret sera						
				B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/Haw 13/04	B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03
B/Shandong/7/1997			2560	320	80	160	2560	<	<	<
B/Tehran/80/2002			1280	320	320	320	2560	<	<	<
B/Brisbane/32/2002			2560	160	80	320	2560	<	<	<
B/Hawaii/13/2004			1280	160	160	160	2560	<	<	<
B/Sichuan/379/1999			<	<	<	<	<	320	640	<
B/Shanghai/361/2002			<	<	<	<	<	320	640	<
B/Jiangsu/10/2003			<	<	<	<	<	80	80	320
B/Christchurch/27/2004	27.8.04	E3 \1	<	<	<	<	<	160	160	<
B/Christchurch/33/2004	30.8.04	E3 \1	<	<	<	<	<	80	80	<
B/Victoria/508/2004	10.9.04	E2 \1	<	<	<	<	<	80	160	<
B/Brisbane/4/2004	11.9.04	E2 \2	<	<	<	<	<	80	160	<
B/Brisbane/5/2004	23.9.04	E3 \1	<	<	<	<	<	80	160	<
B/Victoria/509/2004	23.9.04	E4 \1	<	<	<	<	<	320	640	<
B/Brisbane/13/2004	28.9.04	E2 \1	<	<	<	<	<	80	160	<
B/Iceland/9/2004	17.12.04	MDCK1 \1	<	<	<	<	<	80	160	160
B/Switzerland/5303/2004	14.12.04	MDCK1 \1	2560	80	<	160	80	<	<	<

¹ <, <40

² hyperimmune sheep serum

Table 6D. Antigenic analyses of influenza B viruses (25.1.05)

Viruses	Isolation date		Haemagglutination inhibition titre ¹						
			B/Shan ² 7/97	Post infection ferret sera					
				B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03
B/Shandong/7/1997			2560	320	80	160	<	<	<
B/Tehran/80/2002			1280	320	320	160	<	<	<
B/Brisbane/32/2002			1280	160	80	160	<	<	<
B/Sichuan/379/1999			<	<	<	<	320	640	<
B/Shanghai/361/2002			<	<	<	<	320	640	40
B/Jiangsu/10/2003			<	<	<	<	80	80	320
B/Dakar/81/2004	5.11.04	MDCK2 \1	<	<	<	<	320	320	320
B/Dakar/85/2004	10.11.04	MDCK2 \1	<	<	<	<	320	320	320
B/Dakar/86/2004	10.11.04	MDCK2 \1	<	<	<	<	320	320	320
B/Switzerland/5579/2004	21.12.04	MDCK1 \1	<	<	<	<	320	160	320
B/Umea/1/2005	25.12.04	MDCK2 \1	<	<	<	<	160	320	320
B/Norway/40/2004	28.12.04	MDCK1 \1	<	<	<	<	320	320	320
B/Norway/46/2005	4.1.05	MDCK1 \1	<	<	<	<	320	320	320
B/Egypt/144/2005	5.1.05	E1,E2	<	<	<	<	640	640	40
B/Norway/43/2005	6.1.05	MDCK1 \1	<	<	<	<	320	320	320

¹ <, <40

² hyperimmune sheep serum

Table 6E. Antigenic analyses of influenza B viruses (1.2.05)

Viruses			Haemagglutination inhibition titre ¹							
			Isolation date	B/Shan ² 7/97	Post infection ferret sera					
					B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03
B/Shandong/7/1997			2560	640	160	160	<	<	<	
B/Tehran/80/2002			2560	160	160	80	<	<	<	
B/Brisbane/32/2002			2560	320	160	160	<	<	<	
B/Sichuan/379/1999			<	<	<	<	640	640	<	
B/Shanghai/361/2002			<	<	<	<	640	640	<	
B/Jiangsu/10/2003			<	<	<	<	80	80	320	
B/Sachsen/1/2004	22.12.04	MDCKx \1	<	<	<	<	80	160	40	
B/Rheinland-Pfalz/1/2004	Jan 05	MDCKx \1	<	<	<	<	80	160	40	
B/Sachsen/1/2005	3.1.05	MDCKx \1	<	<	<	<	160	160	160	
B/Lisbon/2/2005	8.1.05	MDCK1 \1	<	<	<	<	80	160	160	
B/Lisbon/3/2005	15.1.05	MDCK1 \1	<	<	<	<	80	160	80	
B/Colorado/4/2004	unknown	E2 \1	<	<	<	<	320	320	<	
B/Fujian/430/2004	unknown	E1,C2,E1 \1	<	<	<	<	80	160	<	
B/Phitsanulok/2053/2004	2.10.04	x1C1,C1E1 \1	1280	160	80	40	<	<	<	
B/Switzerland/5580/2004	21.12.04	MDCK1 \1	2560	160	40	<	<	<	<	
B/Bayern/1/2004	23.12.04	MDCKx \1	2560	320	80	<	<	<	<	

¹ <, <40

² hyperimmune sheep serum

Figure 5. Phylogenetic comparison of B HAs

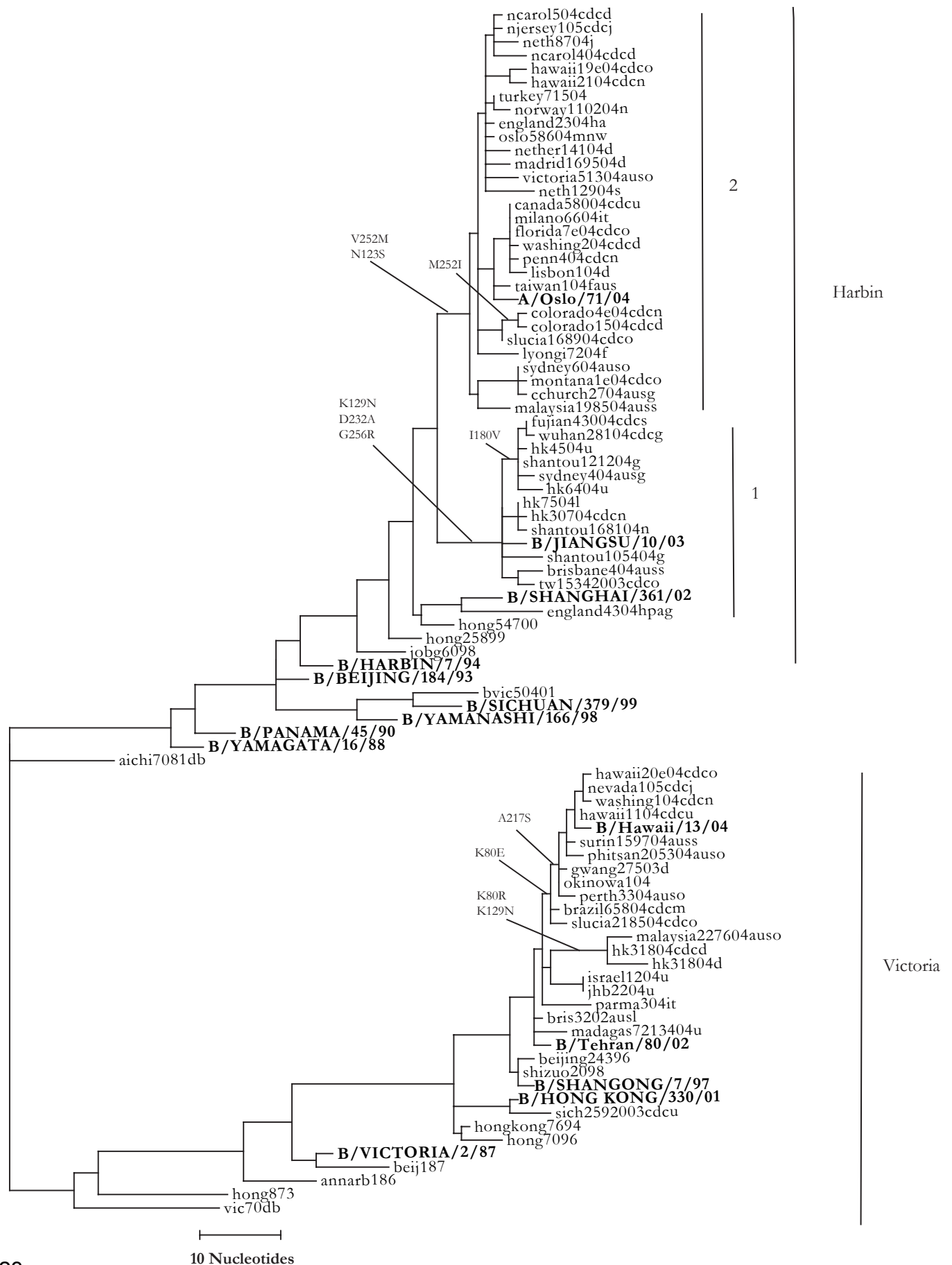


Figure 6. Phylogenetic comparison of BNAs

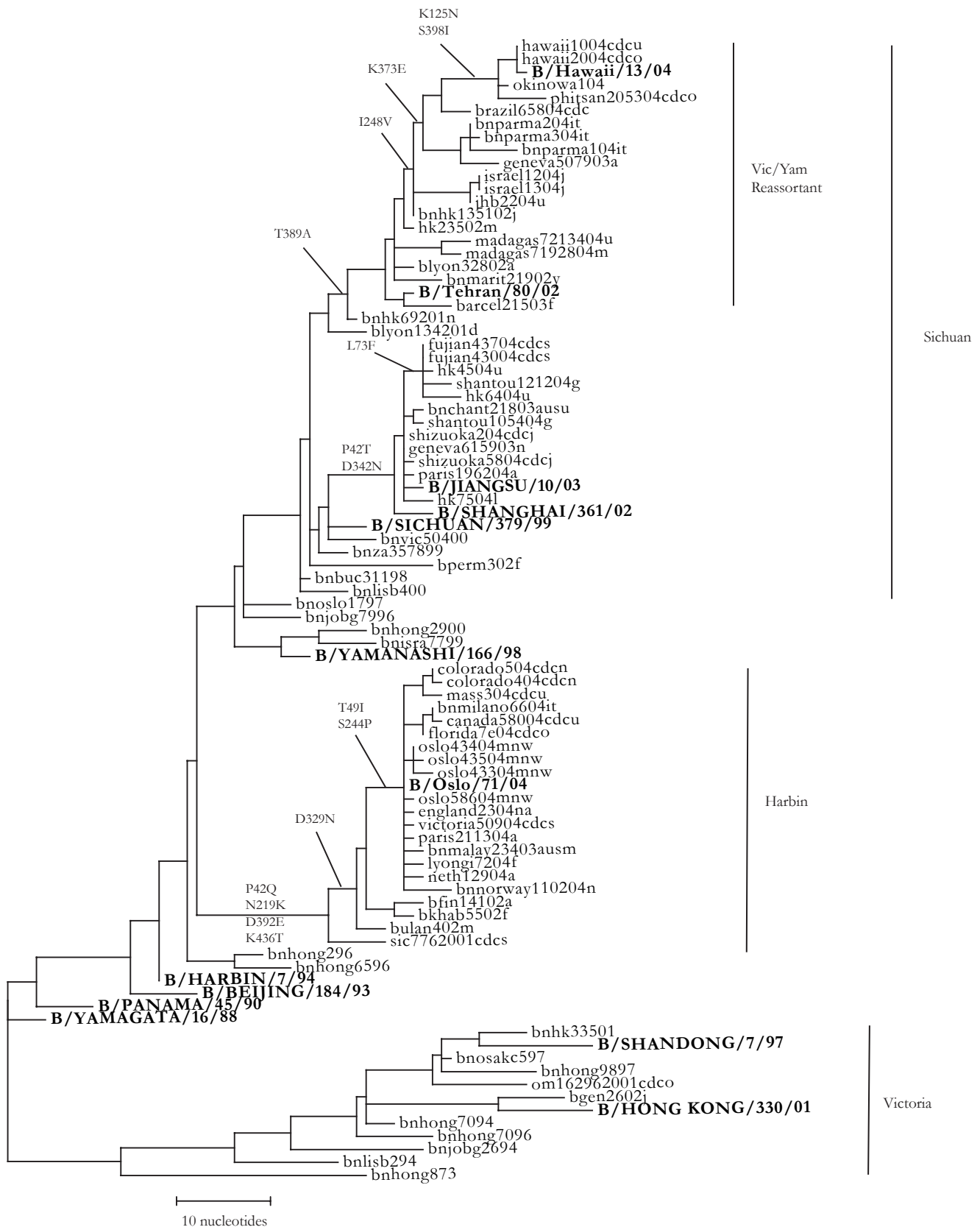


Table 7. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Harbin(1)	B/Jiangsu/10/03	H40Y ¹ K129N D233A G256R (I180V)	
Harbin(2)	B/Oslo/71/04	H40Y ¹ V252M (I) (P108A) (N123S)	T42Q ¹ T49I K186R N219K N235D S244P D329N N342D D392E E436T
Victoria	B/Hawaii/13/04	K80E ² I121T A217S (K299R)	K125N ³ I248V K373E S398I

1. Relative to B/Shanghai/361/02. 2. Relative to B/Shandong/7/97. 3. Relative to B/Tehran/80/02.